

7 (a) (i) Planck's constant. [1]

(ii) For the oil droplets to acquire negative charges, electrons need to be removed from air molecules to attach to them. [1]

**OR**

Air is needed for the oil droplets to experience viscous force to acquire a terminal velocity (for the determination of the droplet's mass). [1]

(b) (i) Reading from the graph,  $W = 5.1 \times 10^{-14}$  N [1]

$$\begin{aligned}\frac{QV}{d} = W & \Rightarrow Q = \frac{Wd}{V} \\ & = \frac{(5.1 \times 10^{-14})(0.00442)}{470} \\ & = 4.80 \times 10^{-19} \text{ C}\end{aligned}$$

[1]

(ii) The density of air is very small (or 1/800) when compared to that of oil. [1]

$$\begin{aligned}\text{(iii)} \quad & \frac{(1.66 + 6.53 + 1.59 + 4.80 + 3.18 + 7.83) \times 10^{-19}}{1 + 4 + 1 + 3 + 2 + 5} \\ & = 1.599 \times 10^{-19} \text{ C (to 4 sig. fig.)}\end{aligned}$$

[1]

(c) (i) Stroboscope / light gates / video camera [1]

(ii) The very small weight (mass) is quickly balanced out by the (effects of) air resistance. [1]

$$\begin{aligned}\text{(iii)} \quad & \text{At terminal velocity, } F_D = W \\ & 6\pi \eta r v = 7.0 \times 10^{-14}\end{aligned}$$

[1]

$$\text{Terminal velocity, } v = \frac{0.00100}{T} = \frac{0.00100}{6.2} = 0.0001613 \text{ m s}^{-1}$$

[1]

$$\begin{aligned}r &= \frac{7.0 \times 10^{-14}}{6\pi (1.8 \times 10^{-5})(0.0001613)} \\ &= 1.28 \times 10^{-6} \text{ m}\end{aligned}$$

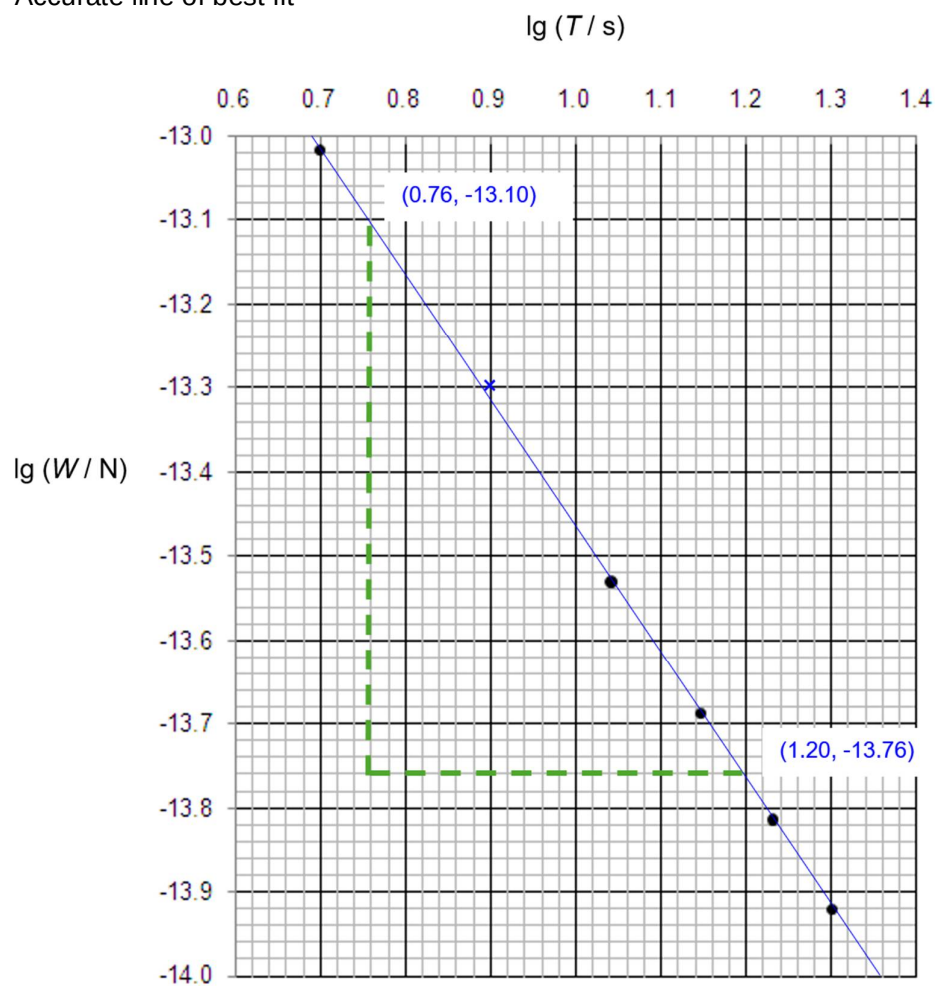
[1]

(d) (i) When  $T = 8.0$  s,  $W = 4.7 \times 10^{-14}$  N.  
 $\lg (W / \text{N}) = -13.33$  [1]

- (ii) Correct plot  
Accurate line of best fit

[1]

[1]



- (iii) Gradient =  $b = \frac{-13.10 - (-13.76)}{0.76 - 1.20}$   
 $= -1.50$  (range: -1.4 to -1.6)

[1]

[1]

- (iv)  $W = aT^{1.5} = a\left(\frac{1}{T}\right)^{1.5}$   
 $v = \frac{d}{T}$ , (where  $d$  is the 1.00 mm fall by the droplet.)

[1]

Since  $a$  and  $d$  are constants,  
 $W \propto v^{1.5}$

[1]